KR-ACC-12 Huawei Access Node Failure SOP (Major)

Node: KR-ACC-12 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_4292

Description

KR-ACC-12 Huawei Access Node Failure SOP (Major) - Introduction. The KR-ACC-12 site, located at KR Puram Access, is experiencing a hardware/environmental fault failure within the Huawei Access Node. This failure is manifesting as ASIC memory parity errors and line card crashes, compounded by the loss of power supply redundancy. The alarm code ALARM_4292 has been triggered, indicating a Major failure within the access node. This SOP outlines the diagnostic and resolution steps necessary to rectify this failure, leveraging the Huawei CLI dialect. The underlying protocol mechanics driving this failure domain revolve around the Line Card's (LC) ability to process and forward traffic, reliant on the integrity of the ASIC's memory and the power supply redundancy.

Pre-requisite Checks

- Check: Verify redundant link status, Description: Ensure redundant link is operational and no other site is affected., Action: Run 'display interface brief' to verify link status

- Check: Check environmental conditions, Description: Verify no overheating, moisture, or other environmental anomalies are present., Action: Review site environmental monitoring data

- Check: Review power supply redundancy, Description: Verify power supply redundancy is operational and no other sites are affected., Action: Run 'display power supply' to verify redundancy status

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display hardware, Expected Output: Hardware information including ASIC memory and line card status, Description: The 'display hardware' command provides real-time information about the system's hardware, including ASIC memory and line card status, which can help identify the root cause of the failure.

Command: display power supply, Expected Output: Power supply status including redundancy information, Description: The 'display power supply' command provides real-time information about the power supply status, including redundancy information, which can help identify if the power supply redundancy loss is the root cause of the failure.

Command: display interface brief, Expected Output: Interface status including link status and errors, Description: The 'display interface
```

```
brief' command provides real-time information about the interface status, including link status and errors, which can help identify if the line card crash is the root cause of the failure.
```

Resolution Steps

Step 1: Step: Reboot the line card, Command: restart line-card 1/1/1, Description: Rebooting the line card will reset the ASIC memory and may resolve the parity error issue.

Step 2: Step: Replace the power supply, Command: replace power-supply 1/1/1, Description: Replacing the power supply will ensure that the power supply redundancy is restored and may resolve the power supply redundancy loss issue.

Step 3: Step: Replace the line card, Command: replace line-card 1/1/1, Description: Replacing the line card will replace the faulty line card and may resolve the line card crash issue.

Post-Check Verification

```
Command: display hardware, Expected Output: Hardware information including ASIC memory and line card status, Description: Verify that the hardware information is correct and the ASIC memory parity error has been resolved.

Command: display power supply, Expected Output: Power supply status including redundancy information, Description: Verify that the power supply status is correct and the power supply redundancy has been restored.

Command: display interface brief, Expected Output: Interface status including link status and errors, Description: Verify that the interface status is correct and the line card crash has been resolved.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: load configuration 1/1/1, Description: Restore the original configuration to ensure that the system reverts back to the previous state.

- Step: Replace the faulty component, Command: replace component 1/1/1, Description: Replace the faulty component to ensure that the system is back to its original state.